



Decay rates of above- and below-ground coarse woody debris of common tree species in New Zealand's natural forest

Loretta G. Garrett^a, Mark O. Kimberley^{a,*}, Graeme R. Oliver^a, Mallory Parks^b, Stephen H. Pearce^a, Peter N. Beets^a, Thomas S.H. Paul^a

^a Scion, Private Bag 3020, Rotorua 3046, New Zealand

^b University of Toronto, 33 Willcocks Street, Toronto, ON M5S 3B3, Canada



ARTICLE INFO

Keywords:

Decomposition
Decay rate
Coarse woody debris
Roots
Dead wood
Temperature

ABSTRACT

Decomposition rates of above-ground coarse woody debris (CWD) (> 10 cm diameter) were determined for nine common New Zealand indigenous tree species, and of below-ground CWD for two species. CWD samples were obtained from fallen trees originating from known windfall events which had been decaying for 10–29 years on the forest floor. Species-specific decay rates were estimated from the density (dry mass divided by original volume) of CWD samples relative to the density of samples from live and freshly fallen trees. The estimated half-life ($t_{1/2}$) or time to lose 50% mass for above-ground CWD across all species was 25.7 years but varied between species and increased with increasing stem diameter. Adjusted to a common diameter at breast height of 60 cm and mean annual temperature of 10 °C, *Prunnopitys taxifolia* and *Fuscospora cliffortioides* were the slowest to decay with a $t_{1/2}$ of about 40 years, *Fuscospora fusca*, *Weinmannia racemosa*, *Beilschmiedia tawa*, *Lophozonia menziesii* and *Prunnopitys ferruginea* the next slowest with a $t_{1/2}$ of 27–34 years, while the remaining species (*Dacrydium cupressinum* and *Dacrycarpus dacrydioides*) decaying more rapidly with a $t_{1/2}$ of less than 20 years. A model was derived for predicting the exponential decay constant for the above-ground CWD of individual stems as a function of diameter at breast height and mean annual temperature adjusted for species. This model can be used to predict above-ground CWD carbon stock changes in New Zealand's natural forests following tree mortality. Below-ground CWD decay rates did not differ significantly from those of above-ground CWD for *D. cupressinum*, but were significantly faster for *L. menziesii*.

1. Introduction

Coarse woody debris (CWD) defined here as material from fallen and standing dead trees greater than 10 cm in diameter, is an important structural and functional component of forest ecosystems and can comprise a significant proportion of the total live plus dead organic matter pools (Harmon et al., 1986). Old growth forests such as New Zealand's natural forests contain large amounts of CWD compared with forests managed for timber production (Ranius et al., 2004). Large inputs into the dead wood pool occur periodically from tectonic activity or following windthrow of large diameter trees or crown breakage during severe storm events. In addition, regular inputs occur as a result of natural mortality of suppressed trees within overstocked stands or of palatable species following excessive browsing. While mortality is the process that creates woody detritus, decomposition is the fundamental process that regulates its loss (Harmon, 2002). The decay rate of woody debris therefore plays a major role in determining the level of carbon

stocks stored in forests in the form of dead wood.

Various factors can affect the decomposition rate of dead wood, which includes standing debris, fallen debris that is either suspended above the ground, in ground contact, or partially buried, and roots that are within the ground or uplifted roots either with or without attached soil. Species differences in decomposition can be considerable with gymnosperm species often decomposing more slowly than angiosperms, possibly due to their higher lignin concentration and lower concentrations of nitrogen and phosphorus (Weedon et al., 2009). More importantly however, are marked individual species-differences in heartwood extractives content, which may contain chemicals that are toxic to wood boring insects (Clifton, 1994) and highly resistant to decay. In contrast to heartwood, sapwood is permeable to water with New Zealand's indigenous species decaying within two years when in ground contact, with the sapwood/heartwood ratio influencing the rate of decay (Clifton, 1994). Environmental drivers are also important. Decomposition rates are positively related to temperature (Chambers

* Corresponding author.

E-mail address: mark.kimberley@scionresearch.com (M.O. Kimberley).

<https://doi.org/10.1016/j.foreco.2018.12.013>

Received 30 June 2018; Received in revised form 5 December 2018; Accepted 7 December 2018

Available online 14 February 2019

0378-1127/ © 2018 Published by Elsevier B.V.

et al., 2000; Russell et al., 2015), and negatively related to piece diameter (Zell et al., 2009). An analysis of a global data set by Mackensen et al. (2003) found mean annual temperature was the main driver of decomposition, with log diameter also influencing the rate of decay. The decrease in sapwood ratio with increasing diameter explains why decay rates decrease with increasing stem diameter within a species. Moreover, factors that determine whether CWD can become dry or alternatively is completely saturated (as in wetlands) will have an overriding influence on decay rates because decay fungi require access to oxygen and water in order to be active.

Decay functions are parameterised using estimates of the initial density of material of a given volume and the density of this same material after decaying for a known period of time (Beets et al., 2008). The exponential decay model of Olson (1963) is by far the most common model used to quantify mass loss, although more complex sigmoidal functions have sometimes been found more appropriate (Fraver et al., 2013). The two primary methods used for the field measurement of decay rates are the time series method where the same pieces are measured repeatedly over time, and the chronosequence method where information is assembled from different sets of pieces which have decayed for different but known lengths of time (Russell et al., 2015). When using the chronosequence method, it is important to take account of log volume loss due to changes in size and shape as decomposition progresses. Failure to do so by using loss in wood density as a surrogate for loss of biomass will lead to an underestimate of the true decay rate, although this source of bias is often ignored (Laiho and Prescott, 2004).

Although most forest CWD is above ground, a significant proportion is found in the roots of dead trees with root/shoot ratios typically averaging 20–25% (Cairns et al., 1997; Beets et al., 2007). However, decay rates of below-ground dead wood have been much less studied than those of above-ground dead wood. A number of studies have reported root decay rates to be similar to or slower than those of above-ground dead wood (Fahey et al., 1988, 1991; Janisch et al., 2005; Olajuyigbe et al., 2011). In New Zealand, root decay studies have only been reported for *Pinus radiata* D. Don where root decay rates were found to be similar to or faster than decay rates of above-ground dead stems (Garrett et al., 2008, 2010, 2012).

New Zealand has agreed to report on carbon exchange in vegetation and soil under the United Nations Framework Convention on Climate Change. This requires estimation of carbon in natural forest, which consists of land classified as forest in 1990 excluding planted forest, consisting predominantly of tall indigenous forest and covering approximately 7.8 million ha (Anon, 2012). A network of permanent plots was established and measured for the first time in 2002–2007 to provide national estimates of carbon stocks in live and dead organic matter pools in natural forest in New Zealand, and to determine whether the forest is a carbon source or sink (Payton et al., 2004). Above-ground CWD is directly measured in each plot and includes all standing dead trees, spars, fallen logs, and stumps that meet the size threshold

(> 10 cm in diameter and 0.3 m in length). Based on this measurement, the above-ground CWD pool in New Zealand's natural forest, is estimated to average 43 Mg ha⁻¹ (Holdaway et al., 2016) or 22 Mg C ha⁻¹ assuming 50% of the biomass is carbon.

Using the plot network, it will be possible to model carbon stock change in the CWD pool by quantifying additions to the pool from tree mortality and modelling losses from decay. This will require the use of decay functions for the major indigenous tree species. Exponential decay functions and associated decay constants for above-ground CWD were reported for seven New Zealand indigenous tree species by Beets et al. (2008). Significant differences in decay rates were found between species along with a significant piece size effect, with fallen stems of larger diameter decaying more slowly than those of smaller diameter. For these seven species the average time to lose 50% mass or half-life ($t_{1/2}$) was 30 years.

However, the decay functions presented by Beets et al. (2008) have a number of limitations when used for predicting decay rates of CWD sampled over the permanent plot network. They include only four of the six most abundant species nationally, and do not include *Weinmannia racemosa* L.f. (kamahi), and *Fuscospora cliffortioides* (Hook f.) Heenan & Smitten (2013) (mountain beech), which are respectively the third and fourth most abundant species nationally (Beets et al., 2009). Also, Beets et al. (2008) only presented decay constants for above-ground CWD, and when applying their model it is therefore necessary to assume above- and below-ground decay rates do not differ.

The objective of this study was therefore firstly to derive decay rates for nine common New Zealand indigenous tree species including two species not covered in the earlier study. A second objective was to test the hypothesis that decay rates of below-ground CWD do not differ from those of above-ground CWD for two major indigenous New Zealand tree species, one a softwood and the other a hardwood.

2. Materials and methods

2.1. Study sites and species

The data used in this study consist of the original dataset of Beets et al. (2008) supplemented with additional measurements, increasing both the sample size per species, the number of species, and the number of sites. The seven species sampled previously at two sites by Beets et al. (2008) were the softwoods *Dacrydium cupressinum* Lamb. (rimu), *Prumnopitys taxifolia* (Sol. ex D. Don) de Laub. (matai), *Prumnopitys ferruginea* (D. Don) de Laub. (miro), and *Dacrycarpus dacrydioides* (A. Rich.) de Laub. (kahikatea), and the hardwoods *Fuscospora fusca* (Hook.f.) Heenan & Smitten (2013) (red beech), *Beilschmiedia tawa* (A. Cunn.) Benth. & Hook.f. ex Kirk (tawa) and *Lophozonia menziesii* (Hook.f.) Heenan & Smitten (2013) (silver beech). Three new natural forest sites in the North Island of New Zealand with elevation varying from 300 m to 1000 m, and two new hardwood species, *W. racemosa*, and *F. cliffortioides*, were added in the current study (Table 1). In

Table 1
Study sites showing mean annual rainfall, mean annual temperature (MAT), elevation, and tree species sampled.

Location	Latitude	Longitude	Rainfall (mm y ⁻¹) ^a	MAT (°C) ^a	Elevation (m.a.s.l.)	Species
Okataina	38°06'S	176°26'E	2064	11.2	514	<i>W. racemosa</i>
Pirongia	37°59'S	175°01'E	2052	12.5	313	<i>W. racemosa</i>
Ruapehu	39°11'S	175°31'E	2712	8.0	1029	<i>F. cliffortioides</i>
Whirinaki ^b	38°39'S	176°42'E	1500	11.0	460	<i>P. ferruginea</i> <i>D. cupressinum</i> <i>B. tawa</i> <i>D. dacrydioides</i> <i>P. taxifolia</i>
Kaimanawa ^b	38°57'S	176°13'E	1700	9.5	700–800	<i>F. fusca</i> <i>L. menziesii</i>

^a Annual rainfall and Mean Annual Temperature (MAT) from Leathwick et al. (2003).

^b Sites described in Beets et al. (2008).

addition, roots were sampled from *D. cupressinum* in the Whirinaki forest and *L. menziesii* in the Kaimanawa forest.

2.2. Sampling procedures

The mean density of live trees of each species was determined using a two-stage sampling procedure described in Beets et al. (2008). Firstly, breast height outerwood cores were obtained from 20 live trees per species (40 trees for *W. racemosa*), and basic density of wood 5–15 cm under bark determined using the maximum moisture content method (Smith, 1954). Basic density is the oven dry weight at 104 °C divided by the green (saturated) volume of wood samples (Clifton, 1994). Note that the outer 5 cm of sapwood was not used because pinhole borer was evident in some of the recently fallen dead trees, and to maintain consistency with Beets et al. (2008). Secondly, the volumes and densities of recently fallen dead trees (including the stem wood, stem bark and large limbs > 10 cm in diameter) were measured as described in the following section, and the breast height outerwood basic density of each of these trees obtained from a wood strip sample. Recent windfall trees with green foliage attached were preferred. In cases where foliage was no longer present, careful inspection was necessary to ensure that wood decay was largely absent. Stem diameter at breast height (dbh) and total tree height were obtained for all trees (including live standing and fallen dead trees) in the study.

Above-ground CWD samples were obtained from windthrown dead trees that had been decaying for 10–29 years (average 23 years) on the forest floor as described in the following section. Windfall trees from cyclone Bernie (in 1982) were located in existing long-term trials at Whirinaki, Kaimanawa, and Ruapehu (Martin, 2006; Beets et al., 2008). At Okataina and Pirongia, National Vegetation Survey (NVS) plots with repeat measurements were used to determine the approximate date when *W. racemosa* windfall had occurred. The above-ground CWD was predominantly in ground contact, and was usually shaded by an overstorey of canopy trees.

Sample sizes and mean dimensions of live trees selected for the outerwood density survey, of recently fallen trees, and of decayed fallen trees, and the number of years of decay, are summarised in Table 2. Standing dead trees were not sampled. Tree measurements, density data and decay times are included as Supplementary Material accompanying this paper.

2.3. Measurement of fallen stems

The procedures used for measuring fallen stems described in (Beets et al., 2008) were followed for the recently fallen and decayed trees in the current study. In summary, for each stem, diameter over bark was measured using a diameter tape, at 0.15 m from base of stem, 0.7 m,

1.4 m (dbh), and depending on stem length, either at 2.0 m or 3.0 m and thereafter at 2.0 or 3.0 m intervals, giving approximately 8–10 measurement points along the main stem up to where the CWD threshold diameter of 10 cm was reached. Diameters of all stem forks (that met the diameter threshold for CWD) were also measured as described for the main stem and included in stem volume calculations. Care was taken to determine the original diameter by making allowances for any bark and sapwood loss that had occurred due to decay and fragmentation. This ensured that the original volume was determined as accurately as possible. At each main stem measurement height, a disc approximately 5 cm in length was cut and its actual length measured at 6–10 points around the circumference of the disc using calipers to determine disc volume. Total stem height and the height to each diameter measurement point up to a small end diameter of 10 cm were recorded. The lengths and large and small end diameters of all branches ≥ 10 cm diameter were measured and discs cut at intervals and measured to determine disc volume, as described for stems. If hollowing was present and considered to be derived from decay prior to tree mortality, the relevant area was deducted from the disc total cross-sectional area to calculate disc volume. After completion of field measurements, the disc samples were oven dried at 70 °C to constant weight. The density of each disc was calculated by dividing the dry weight of the disc by the disc total volume.

2.4. Measurement of roots

Roots were excavated from two fallen *D. cupressinum* and two fallen *L. menziesii* trees, both after 27 years of decay, and from three live *D. cupressinum* trees and two fallen *L. menziesii* trees after 3 years of decay. For each tree, sections of lateral roots were collected from the stump down to a diameter of 10 cm, covering a range of size classes exhibited by the root taper of the particular root plate sampled. Roots were mostly sampled from within the soil but exposed roots were also sampled where the root plate was uplifted due to a windfall event. Buried roots were carefully excavated to ensure digging did not cause breakage of the decaying material and care was taken to remove soil from samples. Between 1 and 6 (mean 2.3) samples were collected from each tree. Each sample was between 5 and 27 cm (mean 10 cm) in length and ranged in diameter from 10 cm to 25 cm (mean 14 cm). Diameter over bark was measured at both ends of the sample and at the mid-point on longer samples, and volume calculated as the sum of one or two truncated cones.

Root samples were weighed individually after oven drying at 70 °C to constant weight. Samples were then bulked per tree, chipped and ground (to < 2 mm), and mass loss on ignition (LOI) determined at 550 °C. The ash remaining after ignition of clean live roots averaged 2.0% (S.E. = 0.2%), indicating little contamination by mineral soil,

Table 2

Species sample sizes (n), mean diameter at breast height (dbh) and height (ht) of live density survey trees, recently fallen trees, decayed fallen trees, and trees used for root sampling, and mean decay time of decayed fallen trees.

Species	Live survey trees			Recently fallen trees			Decayed fallen trees			
	n	dbh (cm)	ht (m)	N	dbh (cm)	ht (m)	n	dbh (cm)	ht (m)	Decay time (yrs)
<i>B. tawa</i>	20	43	28	5	32	24	6	45	29	22
<i>D. cupressinum</i>	20	117	51	10	102	49	8	92	51	20
<i>D. dacrydioides</i>	20	121	49	1	142	54	1	138	50	25
<i>F. cliffortioides</i>	20	38	15	2	30	12	4	28	11	27
<i>F. fusca</i>	20	61	27	2	25	17	14	42	25	23
<i>L. menziesii</i>	20	71	26	2	87	29	9	54	27	23
<i>P. ferruginea</i>	20	54	34	6	51	35	4	42	32	27
<i>P. taxifolia</i>	20	82	41	4	61	37	8	83	44	20
<i>W. racemosa</i>	40	28	18	–	–	–	4	30	16	24
<i>D. cupressinum</i> (root trees)	–	–	–	3	85	48	2	110	49	27
<i>L. menziesii</i> (root trees)	–	–	–	2	68	25	2	66	30	27
Average (excluding root trees)	200	64	31	32	67	36	58	56	31	23

while the ash content of decayed samples averaged 6.2% (SE = 1.2%), indicating some contamination. Sample oven dry weights were expressed on an ash free basis. The mean density of root samples was calculated per tree by dividing the sum of the sample oven dry weights (ash-free) by the sum of the fresh sample volumes.

2.5. Statistical analysis

Species means of whole stem and branch (> 10 cm diameter) density, breast height outer wood density, the ratio of stem and branch to breast height outer wood density, and root density in recently fallen trees were compared using analysis of variance (ANOVA) and the least significant difference (LSD) test, performed using the SAS (Version 9.4) procedure GLM. ANOVAs were also used to compare species means of stem and branch density and root density in decayed fallen trees.

Exponential decay functions for predicting density as a function of decay time were fitted as nonlinear mixed models using the SAS procedure NLMIXED. The following model form was used:

$$D_{ijk} = DS_i(S_{ijk} + O_{ijk}/a + b_i R_{ijk}) \exp(-\lambda_i(S_{ijk} + c_i R_{ijk})t_{ijk}) + tr_{ij} + e_{ijk} \quad (1)$$

In this model, D_{ijk} (kg m^{-3}) is the measured density of the j^{th} tree of the i^{th} species. The subscript k denotes whether the density measurement is for the stem and branch ($k = 1$), breast height outer wood ($k = 2$), or root ($k = 3$). S_{ijk} , O_{ijk} , and R_{ijk} are dummy variables indicating the type of measurement, assigned values of one for stem, outer wood, and root density respectively, and zero otherwise. The time of decay in years is t_{ijk} (with $t_{ijk} = 0$ for live trees). The fitted model parameters are DS_i , a , b_i , c_i and λ_i . DS_i is stem and branch (> 10 cm diameter) density for the i^{th} species, a is the ratio of stem and branch to outer wood density in live trees which following Beets et al. (2008) is assumed not to vary between species, b_i is the ratio of root to stem density in live trees for the i^{th} species, c_i is the ratio of the decay constant for roots to the decay constant for stems and branches for the i^{th} species, and λ_i is the exponential stem and branch decay constant for the i^{th} species. The model has a hierarchical error structure with random error terms tr_{ij} for the j^{th} tree, and e_{ijk} for each individual measurement. Error terms are assumed identically, independently, and normally distributed with mean zero.

Model (1) provided estimates of the above-ground CWD decay constants λ_i for each species. The half-life ($t_{1/2i}$) for density loss was estimated using:

$$t_{1/2i} = \ln(2)/\lambda_i \quad (2)$$

with standard error obtained using the delta method (Oehlert, 1992):

$$\text{se}(t_{1/2i}) = \text{se}(\lambda_i) \ln(2)/\lambda_i^2 \quad (3)$$

As noted by Beets et al. (2008) there was a tendency for density of both live and dead trees to be related to dbh, and because live and dead trees differed in mean dbh it was desirable to also incorporate tree size effects into the decay constant. Model (1) was therefore modified to allow for relationships between initial density and dbh, and decay rate and dbh. Inspection of the dataset indicated that the density/dbh relationship in live trees varied between species, being negative for most species but positive for *W. racemosa*. Therefore, unlike Beets et al. (2008) who used a general model to relate dbh to live density, species-specific relationships were used in this study using:

$$DS_i = DS_{60i}(dbh_{ij}/60)^{d_i} \quad (4)$$

For simplicity, dbh is standardised in Eq. (4) to 60 cm, meaning that the parameter DS_{60i} is the live whole stem density of a 60 cm dbh tree for the i^{th} species, while the parameter d_i accounts for the relationship with dbh for species i .

Secondly, the decay rate was expressed as a function of dbh although in this case, like Beets et al. (2008), a universal relationship across all species was assumed. To achieve this, Model (1) was modified

by expressing the λ_i parameters as a function of dbh using:

$$\lambda_i = \lambda_{60i} + f \ln(dbh_{ij}/60) \quad (5)$$

as used by Beets et al. (2008). Experience using the model of Beets et al. (2008) indicates that although it works well over the range of diameters in the dataset, it does not extrapolate well to larger diameter stems which are occasionally encountered in practice. For example, it predicts negative decay constants (i.e., the decay rate is effectively undefined) for stems greater than 3.3 m dbh. Therefore, the following alternative model with the property that $t_{1/2}$ increases linearly with dbh was tested:

$$\lambda_i = \lambda_{60i}/(1 + f(dbh_{ij} - 60)) \quad (6)$$

Note that in both Eqs. (5) and (6), λ_{60i} is the decay constant for a tree of 60 cm dbh in species i , while the parameter f accounts for the relationship between decay rate and dbh. Aikake's Information Criteria (AIC) was used to compare the simpler Model (1) and the more complex models which included dbh in their formulation.

As temperature varies widely with both latitude and elevation within New Zealand indigenous forests, and is known to affect decay rates, it would be useful to adjust the species-specific decay constants derived using the above procedures to account for temperature. However, the range of temperatures in the study was insufficient to allow this effect to be quantified, and it was therefore decided to adjust the decay constants based on the assumption that λ varies in proportion to $\exp(0.093 \times MAT)$ as established by Mackensen et al. (2003) based on a global database, where MAT is mean annual temperature ($^{\circ}\text{C}$). Decay constants, $\lambda_{60,10i}$ adjusted to a common MAT of 10°C and a dbh of 60 cm were calculated using:

$$\lambda_{60,10} = \lambda_{60}/\exp(0.093 \times (MAT - 10)) \quad (7)$$

where MAT is mean annual temperature ($^{\circ}\text{C}$) of the experimental site in the study at which the decay constant was derived.

3. Results

A total of 32 recently fallen trees and a further 58 decayed fallen trees were sampled to determine respectively the initial stem and branch (> 10 cm diameter) density and density following a period of decay after a dated windfall event (Table 2). Initial stem and branch density varied between species (Table 3) with the highest density species, *P. taxifolia*, being 42% greater than the lowest density species, *D. dacrydioides*. Initial stem and branch density decreased with increasing dbh for most species as reported by Beets et al. (2008), but increased with dbh for *W. racemosa*. In contrast, stem and branch density of decayed fallen trees tended to increase with increasing dbh, reflecting the slower decay rate of large stems (Beets et al., 2008). The ratio of stem and branch density to breast height outer wood basic density did not vary significantly between species ($F_{7,23} = 0.51$, $p = 0.82$) and averaged 0.909 (Table 3), verifying the assumption in Beets et al. (2008) that a common ratio could be used for all species. The decay constants for each species estimated using Model (1), and the corresponding half-lives, are shown in Table 4.

A significant improvement in the fit of Model (1) was achieved by including dbh in the model formulation. The best fitting model used Eq. (4) for initial density and Eq. (6) for the decay constant, and reduced the AIC compared with Model (1) by 43 units. Parameter a provides an estimate of stem and branch (> 10 cm diameter) density to breast height outer wood density in recently fallen stems and was 0.912 (s.e. 0.011) which differs only slightly from the simple mean for this ratio of 0.909 shown in Table 2. Parameter f in Eq. (6), which accounts for the effect of dbh on decay rate, was 0.00908 (s.e. 0.00177) and was statistically significant ($t = 5.14$, $p < 0.0001$). Estimates of decay constants from the model along with $t_{1/2}$ for each species adjusted to a dbh of 60 cm are shown in Table 4 along with decay constants further adjusted to a MAT of 10°C using Eq. (7). These tabulated decay constants

Table 3

Mean and standard error by species of breast height outer wood basic density (DOW) of live and recently fallen trees, density of stem and branch (> 10 cm diameter) (DS) of recently fallen and decayed fallen trees, and the ratio of stem and branch density to breast height outer wood density of recently fallen trees. Also shown are densities of root samples (> 10 cm diameter) in recently fallen and in decayed trees. Values in a column followed by the same letter do not differ significantly (LSD test; $\alpha = 0.05$).

Species	DOW of live and recently fallen trees (kg m ⁻³)			DS of recently fallen trees (kg m ⁻³)			DS of decayed fallen trees (kg m ⁻³)			Ratio DS/DOW of recently fallen trees		
	Mean	s.e.		Mean	s.e.		Mean	s.e.		Mean	s.e.	
<i>B. tawa</i>	545	a	8	502	ab	14	250	bc	21	0.901	a	0.034
<i>D. cupressinum</i>	462	f	8	434	cd	10	226	bc	18	0.939	a	0.021
<i>D. dacrydioides</i>	386	g	9	273	f	32	160	bc	52	0.845	a	0.067
<i>F. cliffortioides</i>	485	de	9	480	abc	23	270	b	26	0.904	a	0.048
<i>F. fusca</i>	510	bc	9	482	ab	23	267	b	14	0.880	a	0.048
<i>L. menziesii</i>	498	cd	9	487	ab	23	255	b	17	0.889	a	0.048
<i>P. ferruginea</i>	530	ab	8	473	bc	13	185	bc	26	0.899	a	0.027
<i>P. taxifolia</i>	550	a	8	525	a	16	367	a	18	0.904	a	0.034
<i>W. racemosa</i>	473	ef	7	–	–	–	189	bc	26	–	–	–
<i>D. cupressinum</i> (roots)				411	de	18	207	bc	36			
<i>L. menziesii</i> (roots)				383	e	23	110	c	36			
Average (excluding roots)	492		4	467		10	259		9	0.909		0.011

$\lambda_{60,10}$ can be used to estimate the decay constant λ for a stem of known dbh at a site with a given MAT using the following equation:

$$\lambda = \lambda_{60,10} \times \exp(0.093 \times (MAT - 10)) / (1 + 0.00908 \times (dbh - 60)) \quad (8)$$

The ratios of root (> 10 cm diameter) density to stem and branch (> 10 cm diameter) density in recently fallen trees (parameters b_i in Model (1)) were 0.908 (s.e. 0.054) for *D. cupressinum* and 0.953 (s.e. 0.065) for *L. menziesii*, indicating that densities of roots in recently fallen trees were similar to or slightly lower than the densities of stems and branches for both species. The estimates for the c_i parameters in Model (1) were 0.908 (s.e. 0.16) for *D. cupressinum* and 2.14 (s.e. 0.34) for *L. menziesii*. Note that the test of the hypothesis that the below-ground CWD decay rate equals the above-ground CWD decay rate for species i , is equivalent to testing the hypothesis that $c_i = 1$. This test was statistically significant for *L. menziesii* ($t = 3.33$, $p = 0.00099$) but not for *D. cupressinum* ($t = 0.57$, $p = 0.57$). This indicates that below-ground CWD decayed significantly faster than above-ground CWD for *L. menziesii* (i.e., c_i was significantly greater than one for this species), but not for *D. cupressinum*.

4. Discussion

This study extends the work of Beets et al. (2008) who estimated above-ground CWD decay rates based on the initial density of live trees

and loss in mass due to insect tunnelling and fungal decay approximately 20 years following windfall for seven indigenous New Zealand tree species. The estimated $t_{1/2}$ of above-ground CWD obtained in our study averaged evenly across species was 25.7 years, while the simple average of stems adjusted to a common dbh of 60 cm was 28.8 years, similar to the value of 29.8 years reported in the earlier study. Adjusted to a dbh of 60 cm and a mean annual temperature of 10 °C, decay rates varied greatly between species with *P. taxifolia* (a softwood) and *F. cliffortioides* (a hardwood) being the slowest to decay with a $t_{1/2}$ of about 40 years, *F. fusca*, *W. racemosa*, *B. tawa*, *L. menziesii* and *P. ferruginea* (the only softwood species in this group) next slowest with a $t_{1/2}$ of 27–34 years, while the remaining species, *D. cupressinum*, and *D. dacrydioides* (both softwoods) decay more rapidly with a $t_{1/2}$ of < 20 years. Considering the large variability among species, the decay rates for the New Zealand species measured are broadly comparable with estimates from international studies. For example, an analysis by Russell et al. (2015) of 117 decomposition rates representing 87 genera and/or species, predicts the 25th, 50th and 75th percentiles of the distribution of $t_{1/2}$ to be 9, 15 and 22 years respectively at the average MAT of the New Zealand study sites of 10 °C.

The main controlling factors that influence the decay of wood in a natural environment are debris composition and size, temperature, moisture availability, ground contact, and the decomposer community (Harmon et al., 1986). At the tree level, dbh was shown to be important

Table 4

Stem and branch (> 10 cm diameter) decay constants (λ), half-lives ($t_{1/2}$, years) and standard errors, uncorrected for diameter at breast height (dbh) or mean annual temperature (MAT), adjusted to a standardised dbh of 60 cm, and adjusted to a standardised dbh of 60 cm and MAT of 10 °C, by species. Values in a column followed by the same letter do not differ significantly (LSD test; $\alpha = 0.05$).

Species	Uncorrected for dbh or MAT				Standardised to 60 cm dbh				Standardised to 60 cm dbh and 10 °C MAT			
	λ	s.e.	$t_{1/2}$	s.e.	λ_{60}	s.e.	$t_{1/2}$	s.e.	$\lambda_{60,10}$	s.e.	$t_{1/2}$	s.e.
<i>B. tawa</i>	0.0306	a	0.0035	22.6	2.6	0.0256	cd	0.0029	27.1	3.1	0.0233	29.7
<i>D. cupressinum</i>	0.0279	ab	0.0032	24.9	2.8	0.0385	e	0.0040	18.0	1.9	0.0350	19.8
<i>D. dacrydioides</i>	0.0340	ab	0.0113	20.4	6.8	0.0555	de	0.0178	12.5	4.0	0.0506	13.7
<i>F. cliffortioides</i>	0.0183	bc	0.0031	37.9	6.5	0.0135	b	0.0025	51.4	9.4	0.0162	42.7
<i>F. fusca</i>	0.0240	b	0.0021	28.9	2.6	0.0195	bc	0.0020	35.6	3.6	0.0204	34.0
<i>L. menziesii</i>	0.0249	b	0.0027	27.8	3.0	0.0235	cd	0.0023	29.5	2.9	0.0246	28.2
<i>P. ferruginea</i>	0.0331	a	0.0039	21.0	2.5	0.0285	d	0.0032	24.4	2.7	0.0259	26.7
<i>P. taxifolia</i>	0.0156	c	0.0024	44.5	6.8	0.0188	bc	0.0025	36.9	4.9	0.0171	40.5
<i>W. racemosa</i>	0.0349	a	0.0052	19.8	3.0	0.0261	cd	0.0039	26.5	4.0	0.0220	31.5
<i>D. cupressinum</i> (roots)	0.0267		0.0048	26.0	4.7							
<i>L. menziesii</i> (roots)	0.0516		0.0091	13.4	2.4							
Average evenly weighted by species (excluding roots)	0.0270		0.0016	25.7	1.6	0.0277		0.0022	25.0	2.0	0.0261	26.5
Simple average of stems (excluding roots)	0.0242		0.0016	28.6	1.9	0.0241		0.0012	28.8	1.5	0.0235	29.5

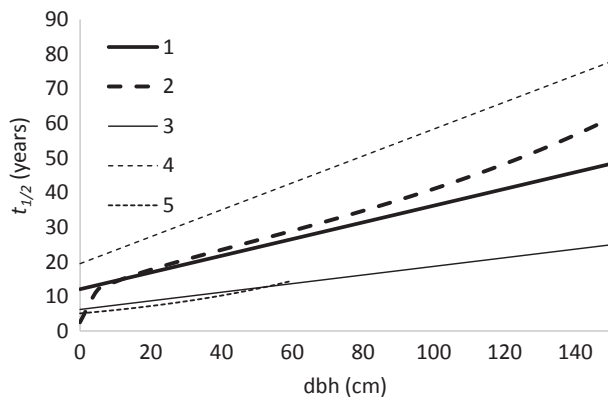


Fig. 1. Effect of tree diameter at breast height (DBH) on time taken to lose 50% of initial mass ($t_{1/2}$) predicted by the following models: (1) average of nine indigenous tree species, current study; (2) average of seven indigenous tree species, Beets et al. (2008); (3) *D. dacrydioides*, current study; (4) *F. cliffortioides*, current study; (5) *P. radiata*, Garrett et al. (2010).

by Beets et al. (2008). In addition, decomposer fungi species and animals can influence decay rates and fragmentation. For example, Beets et al. (2008) found that large stems heavily colonized with the fungi *Ganoderma applanatum* (Pers.) Pat. decayed more rapidly than stems occupied by other fungi. Garrett et al. (2010) found that CWD fragmentation from feral pig (*Sus scrofa* L.) disturbance was associated with increased stem and stump decay, while accelerated decay of woody root sapwood was attributed to activities of the wood boring larvae *Prioporus reticularis* (Huhu grub). The influence of *G. applanatum* on the decay rate of the additional trees added to the original dataset was considered low, with a very low presence of fruiting bodies observed (one or two small brackets) on CWD of one *P. ferruginea*.

A meta-analysis of 36 studies world-wide found that gymnosperms decompose more rapidly than angiosperms within sites (Weedon et al. 2009), although these studies were predominantly based in North American and Western European boreal and temperate forests, which these authors recognised as a data limitation. The results of our multi-species studies in New Zealand are not consistent with their finding, which indicates that other factors can predominate. It is noteworthy that *P. taxifolia* has a low sapwood ratio, *D. dacrydioides* has a high sapwood ratio, and *D. cupressinum* has a wide zone of non-durable transition wood between the sapwood and its durable heartwood, which explains why *D. cupressinum* ranks higher than *P. taxifolia* in a natural durability rating system based entirely on heartwood of indigenous and exotic timber species grown in New Zealand (Clifton, 1994). In this rating system, very durable wood ($t_{1/2} > 25$ years) included both softwood and hardwood species (Clifton, 1994). A comparative study into the natural durability of heartwood and sapwood of a range of tree species in an above-ground “fence-cross test” revealed that very durable heartwood (expected years to failure $t_{1/2} > 30$ years), and in exceptional cases sapwood, was also evident both in softwood and hardwood species, which was ascribed in part to differences in extractives content (Highley, 1995). Unfortunately, species-rankings derived from durability rating systems based on standardised testing conditions do not correlate closely with decay rates obtained for CWD logs under natural forest conditions (Beets et al., 2008), partly because no allowance is made for differences in the sapwood ratio.

Compared with the earlier study of Beets et al. (2008), sample sizes of several species were increased, and data were acquired for two additional species. Our study includes seven species among the most abundant by volume nationally in New Zealand (Beets et al., 2009). These are the beech species, *L. menziesii*, *F. fusca*, and *F. cliffortioides* (ranked first, second, and fourth respectively), the two most abundant non-beech hardwoods, *W. racemosa* and *B. tawa* (ranked third and sixth respectively), and the two most important softwood species *D.*

cupressinum and *P. ferruginea* (ranked fifth and tenth, respectively).

Of the two new species studied, *F. cliffortioides* had one of the slowest decay rates measured to date, while *W. racemosa* because of its small average dbh had one of the fastest. The increased sample size for *P. ferruginea* resulted in an decrease in decay rate (constant 0.0331 yr^{-1} , $t_{1/2} = 21$ years) compared with the result reported by Beets et al. (2008) (0.052 yr^{-1} , $t_{1/2} = 13$ years). However, the decay rate for this species standardised to 60 cm dbh (0.0285 yr^{-1} , $t_{1/2} = 24$ years) was faster than the result reported in the earlier study standardised to 61.9 cm dbh (0.0235 yr^{-1} , $t_{1/2} = 30$ years), emphasising the importance of allowing for tree size effects on decay rates.

This study also adds to the current best available estimates of initial stem and branch (> 10 cm diameter) wood and bark density for a number of species. No significant difference in the ratio of stem and branch density to breast height outer wood basic density was found among species, with this ratio averaging 0.909. This indicates that the mean initial whole stem density for a species can be simply estimated from breast height outer wood density core samples. This is a useful finding, because direct determination of whole stem density is expensive, and can only be obtained infrequently from felled or recently fallen trees in New Zealand where indigenous tree felling is generally not permitted. The models used for estimating decay constants in both this study and the study of Beets et al. (2008) also assume that this ratio does not vary between species.

The effect of above-ground CWD fragmentation on the measurement of density and the potential for an underestimation of the decay constant was considered to be low, as the original stem diameters were measured or determined after allowing for any bark and sapwood loss, following Beets et al. (2008) who tested this approach using time series measurements. Fragmentation is generally considered to be difficult to allow for in chronosequence studies (Lambert et al., 1980; Sollins, 1982). However, even after 27 years of decay, enough bark was still present in the New Zealand tree species studied here to determine the initial over-bark diameter at standard measurement points along the stem, which allowed accurate estimates of the volume and hence density of the above-ground CWD samples.

The effect of tree diameter on $t_{1/2}$ averaged across all species and for the fastest and slowest decaying species is shown in Fig. 1. The model of Beets et al. (2008), and the *P. radiata* model of Garrett et al. (2010) are included in Fig. 1 for comparison purposes. All three models show a similar increase in decay time with increasing diameter although the range of diameters in the *P. radiata* study was narrower than in the indigenous tree studies which included several very large decayed stems over 1 m in diameter. Averaged across all species, $t_{1/2}$ is predicted to be 26.5 years for a 60 cm stem (see Table 4) and 48 years for a 150 cm stem (Fig. 1). When compared with indigenous tree species, the $t_{1/2}$ for plantation-grown *P. radiata* found by Garrett et al. (2010) of 14.5 years is similar to that of *D. dacrydioides*, which when adjusted for dbh is the fastest decaying indigenous species included in our study.

The effect of site mean annual temperature on decay has been reported for various woody debris species by Chambers et al. (2000) and Mackensen et al. (2003) with a Q_{10} temperature coefficient (which quantifies the increased rate of decay corresponding to an increase in temperature of 10°C) of 2.4 and 2.53, respectively. Garrett et al. (2010) reported the effect of mean annual temperature and dbh on *P. radiata* decay in New Zealand based on data from seven locations nationwide ranging in mean temperature from 8 to 12°C . Because the range of temperatures in our study was insufficient to quantify the effect of temperature on decay rate, we have applied the global relationship obtained by Mackensen et al. (2003) to obtain temperature adjusted decay constants for each species (Table 4).

Downed material has been reported to decay more rapidly than standing dead trees and spars, which were not sampled in our study. The downed trees in our study typically had forks and large limbs, not all of which were in ground contact, and hence the species-specific decay rates we obtained using fallen CWD may not have differed greatly

from standing dead trees. We also did not sample buried dead wood. This is defined as material buried more than 50% in the forest floor, and has been reported to decay more slowly than above ground dead wood (Moroni et al., 2015).

The precision of the decay constant estimate for a species is affected by many factors including the number of live trees sampled, the number of decayed CWD trees sampled, and their level of decay. As the number of live trees sampled (20–40 for each species) was generally adequate, the most important factor controlling the precision of estimates in this study was the number of decayed CWD trees sampled. For several species, only 4 decayed trees were sampled (*P. ferruginea*, *W. racemosa* and *F. cliffortioides*) yielding standard errors averaging 15% of the mean value for $t_{1/2}$, while for species with 8–9 trees sampled (*D. cupressinum*, *P. taxifolia* and *L. menziesii*) the standard error averaged 11% of the mean. The most intensively studied species (*F. fusca*), with 14 decayed fallen trees sampled, provided a standard error of 10% of the mean. As a rule of thumb, therefore, it appears that a sample of 15 decayed trees should achieve a standard error of 10% of the mean.

Our study also obtained estimates of below-ground CWD decay rates for two species. The below-ground CWD decay rate for *D. cupressinum* did not differ significantly from above-ground rate, but was significantly faster for *L. menziesii*. Previous studies comparing stem and root decay rates for *P. radiata* in New Zealand indicate that dead root decay rates are similar to or faster than stem decay rates (Garrett et al., 2008, 2010, 2012). Overall the *D. cupressinum* and *L. menziesii* coarse woody root decay was much slower than that of *P. radiata* root decay which had a $t_{1/2}$ of 2–5 years for root systems that were entirely buried within moderately to well-drained soil. Root decay rates observed in our study can be expected to have been influenced by the proportion that decayed within the soil, and on the amount of soil remaining attached to uplifted CWD root material over time. Factors such as these are difficult to determine after 27 years of decay, although clearly of relevance (Moroni et al., 2015).

5. Conclusions

Decomposition rates of above-ground CWD were determined for nine common New Zealand indigenous tree species, and of below-ground CWD for two species. The exponential decay rate constant averaged 0.0270 yr^{-1} ($t_{1/2} = 25.7$ years) across species but differed significantly between species and decreased with increasing stem diameter. A model was derived for predicting the exponential decay constant for individual trees as a function of dbh and MAT adjusted for species. This model can be used to predict above-ground CWD carbon stock changes in New Zealand natural forests following tree mortality. Below-ground CWD decay rates did not differ significantly from those of above-ground CWD for *D. cupressinum*, but were significantly faster for *L. menziesii*.

Acknowledgements

This work was funded by the Foundation for Research Science and Technology (contract CO4X0706) and the Ministry for the Environment (contract 20059). Access to site information and data including the National Vegetation Survey database is greatly appreciated, as is personal communications from Tim Martin and private land owners. Access to study sites was facilitated by Department of Conservation and private land owners.

References

Anon, 2012. New Zealand's greenhouse gas inventory 1990–2012. In: Ministry for the Environment, Wellington, p. 438.
 Beets, P.N., Hood, I.A., Kimberley, M.O., Oliver, G.R., Pearce, S.H., Gardner, J.F., 2008. Coarse woody debris decay rates for seven indigenous tree species in the central North Island of New Zealand. *For. Ecol. Manage.* 256, 548–557.
 Beets, P.N., Kimberley, M.O., Goulding, C.J., Garrett, L.G., Oliver, G.R., Paul, T.S.H.,

2009. Natural forest plot data analysis: carbon stock analyses and re-measurement strategy. Unpublished contract report prepared for the Ministry for the Environment, Wellington.
 Beets, P.N., Pearce, S.H., Oliver, G.R., Clinton, P.W., 2007. Root/shoot ratios for deriving below-ground biomass of *Pinus radiata* stands. *N. Z. J. For. Sci.* 37, 267–288.
 Cairns, M.A., Brown, S., Helmer, E.H., Baumgardner, G.A., 1997. Root biomass allocation in the world's upland forests. *Oecologia* 111, 1–11.
 Chambers, J.Q., Higuchi, N., Schimel, J.P., Ferreira, L.V., Melack, J.M., 2000. Decomposition and carbon cycling of dead trees in tropical forests of the central Amazon. *Oecologia* 122, 380–388.
 Clifton, N.C., 1994. New Zealand Timbers. The Complete Guide to Exotic and Indigenous Woods. GP Publications Ltd.
 Fahey, T.J., Hughes, J.W., Pu, M., Arthur, M.A., 1988. Root decomposition and nutrient flux following whole-tree harvest of northern hardwood forest. *For. Sci.* 34, 744–768.
 Fahey, T.J., Stevens, P.A., Hornung, M., Rowland, P., 1991. Decomposition and nutrient release from logging residue following conventional harvest of sitka spruce in north Wales. *Forestry* 64, 289–301.
 Fraver, S., Milo, A.M., Bradford, J.B., D'Amato, A.W., Kenefic, L., Palik, B.J., Woodall, C.W., Brisette, J., 2013. Woody debris volume depletion through decay: implications for biomass and carbon accounting. *Ecosystems* 16, 1262–1272.
 Garrett, L.G., Kimberley, M.O., Oliver, G.R., Pearce, S.H., Beets, P.N., 2012. Decomposition of coarse woody roots and branches in managed *Pinus radiata* plantations in New Zealand – a time series approach. *For. Ecol. Manage.* 269, 116–123.
 Garrett, L.G., Kimberley, M.O., Oliver, G.R., Pearce, S.H., Paul, T.S.H., 2010. Decomposition of woody debris in managed *Pinus radiata* plantations in New Zealand. *For. Ecol. Manage.* 260, 1389–1398.
 Garrett, L.G., Oliver, G.R., Pearce, S.H., Davis, M.R., 2008. Decomposition of *Pinus radiata* coarse woody debris in New Zealand. *For. Ecol. Manage.* 255, 3839–3845.
 Harmon, M.E., 2002. Moving towards a new paradigm for woody detritus management. In: Proceedings of the Symposium on the Ecology and Management of Dead Wood in Western Forests, General Technical Report PSW-GTR-181. USDA Forest Service, Albany, California, pp. 929–944.
 Harmon, M.E., Franklin, J.F., Swanson, F.J., Sollins, P., Gregory, S.V., Lattin, J.D., Anderson, N.H., Cline, S.P., Aumen, N.G., Sedell, J.R., Lienkaemper, G.W., Cromack, K., Cummins, K.W., 1986. Ecology of coarse woody debris in temperate ecosystems. *Adv. Ecol. Res.* 15, 133–302.
 Highley, T.L., 1995. Comparative durability of untreated wood in use above ground. *Int. Biodeterior. Biodegr.* 409–419.
 Holdaway, R.J., Easdale, T.A., Carswell, F.E., Richardson, S.J., Peltzer, D.A., Mason, N.W.H., Brandon, A.M., Coomes, D.A., 2016. Nationally representative plot network reveals contrasting drivers of net biomass change in secondary and old-growth forests. *Ecosystems* 20, 944–959.
 Janisch, J.E., Harmon, M.E., Chen, H., Fasth, B., Sexton, J., 2005. Decomposition of coarse woody debris originating by clearcutting of an old-growth conifer forest. *Ecoscience* 12, 151–160.
 Laiho, R., Prescott, C.E., 2004. Decay and nutrient dynamics of coarse woody debris in northern coniferous forests: a synthesis. *Can. J. For. Res.* 34, 763–777.
 Lambert, R.L., Lang, G.E., Reinert, W.A., 1980. Loss of mass and chemical change in decaying boles of a subalpine balsam fir forest. *Ecology* 61, 1460–1473.
 Leathwick, J., Wilson, G., Rutledge, D., Wardle, P., Morgan, F., Johnston, K., McLeod, M., Kirkpatrick, R., 2003. Land Environments of New Zealand. Ministry for the Environment/Manaaki Whenua Landcare Research, Wellington/Hamilton.
 Mackensen, J., Bauhus, J., Webber, E., 2003. Decomposition rates of coarse woody debris – a review with particular emphasis on Australian tree species. *Aust. J. Bot.* 51, 27–37.
 Martin, T., 2006. A dendroecological reconstruction of storm events in the forests of central and southern North Island, New Zealand: 1300–2000 AD. Environmental Science. The University of Auckland, Auckland.
 Moroni, M.T., Morris, D.M., Shaw, C., Stokland, J.N., Harmon, M.E., Fenton, N.J., Merganičová, K., Merganič, J., Okabe, K., Hagemann, U., 2015. Buried wood: a common yet poorly documented form of deadwood. *Ecosystems* 18, 605–628.
 Oehlert, G.W., 1992. A note on the delta method. *Am. Stat.* 46, 27–29.
 Olajuyigbe, S.O., Tobin, B., Gardiner, P., Nieuwenhuis, M., 2011. Stocks and decay dynamics of above- and belowground coarse woody debris in managed Sitka spruce forests in Ireland. *For. Ecol. Manage.* 262, 1109–1118.
 Olson, J.S., 1963. Energy storage and the balance of producers and decomposers in ecological systems. *Ecology* 44, 322–331.
 Payton, I.J., Newell, C.L., Beets, P.N., 2004. New Zealand Carbon Monitoring System – Indigenous Forest and Shrubland Data Collection Manual. The Caxton Press, Christchurch, N.Z.
 Ranius, T., Jonsson, B.G., Kruys, N., 2004. Modeling dead wood in Fennoscandian old-growth forests dominated by Norway spruce. *Can. J. For. Res.* 34, 1025–1034.
 Russell, M.B., Fraver, S., Aakala, T., Gove, J.H., Woodall, C.W., D'Amato, A.W., Ducey, M.J., 2015. Quantifying carbon stores and decomposition in dead wood: a review. *For. Ecol. Manage.* 350, 107–128.
 Smith, D.M., 1954. Maximum moisture content method for determining the specific gravity of small wood samples. In: USDA Forest Service, Forest Products Laboratory, Paper No. 2014.
 Sollins, P., 1982. Input and decay of coarse woody debris in coniferous stands in western Oregon and Washington. *Can. J. For. Res.* 12, 18–28.
 Weedon, J.T., Cornwell, W.K., Cornelissen, J.H., Zanne, A.E., Wirth, C., Coomes, D.A., 2009. Global meta-analysis of wood decomposition rates: a role for trait variation among tree species? *Ecol. Lett.* 12, 45–56.
 Zell, J., Kändler, G., Hanewinkel, M., 2009. Predicting constant decay rates of coarse woody debris—a meta-analysis approach with a mixed model. *Ecol. Model.* 220, 904–912.